
CLAM: **Causal Spatial Disaggregation** **to Infer Local Effects From Coarse Data**

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Abstract

Learning fine-grained spatial patterns from coarse-resolution data is challenging, especially in causal settings where high-resolution effects must be inferred from aggregated interventions and outcomes. We introduce CLAM, a method for estimating localized causal effects from coarse observations by exploiting high-resolution contextual covariates that modulate these effects. By jointly learning the causal mechanism and a disaggregation mapping, CLAM captures interactions that are missed when addressing these problems independently. The method supports localized effect estimation, counterfactual reasoning, and principled outcome disaggregation, and reliably captures spatially varying causal effects across diverse settings. This is particularly relevant for applications such as public health and environmental policy, where decisions are made at broad scales despite substantial local heterogeneity. Code is available at github.com/gerritgr/clam.

1 Introduction

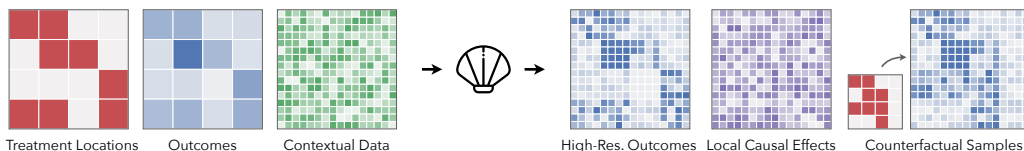


Figure 1: **Motivating example.** CLAM takes regional treatments (e.g., a vaccination awareness campaign) and outcomes (e.g., vaccination levels), along with subregional covariates (e.g., demographics), to learn treatment–outcome relationships and estimate local causal effects or sample counterfactuals.

Going from high-resolution (HR) to low-resolution (LR) data is typically straightforward, as it can be done via aggregation or pooling functions such as sums or means. These operations generally remove information. In contrast, going from LR to HR data is fundamentally difficult because it requires restoring information that has been lost. This generally necessitates some form of prior knowledge about the HR data. This inverse problem is typically studied under terms such as *spatial disaggregation* [1], *downscaling* [2], *ecological inference* [3], or *super-resolution* [4]. It is relevant across many fields involving spatio-temporal statistics, including earth science [5], public health [6], and the social sciences [7].

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Causality [8], on the other hand, deals with the effect of interventions on outcome variables. Its core insight is that statistical associations do not imply causal relationships. In other words, observing a system is not the same as knowing how intervening on it will change its dynamics.

Answering causal questions is often attempted using LR data, but this data is frequently insufficient for this [9]. For example, interventions, like vaccination campaigns, might be applied across broad regions, yet their impact could vary significantly at the subregional level. Surprisingly little effort has been made to properly combine spatial disaggregation with the quest to answer causal questions.

Our hypothesis is that causal inference and spatial disaggregation are better solved together than apart, and that doing so makes questions answerable that neither can address alone.

Contribution. This work addresses the gap between spatial disaggregation and causal inference by introducing a unified framework grounded in structural causal models (SCMs) and Pearl’s do-calculus [8]. We argue that jointly tackling causal inference and disaggregation is natural in many real-world settings, where interventions are implemented at coarse scales but effects manifest locally. While our primary focus is the formulation of this framework, we illustrate its capabilities and limitations through a series of simulation studies that mimic realistic data-generating processes, as well as a case study based on real-world data on the impact of heat waves on gun violence.

CLAM (*Causal disaggregation method*) can be summarized as follows: We learn a mechanistic function that maps an intervention indicator and contextual information to high-resolution (HR) outcomes. In doing so, we treat HR contextual data (or *covariates*) as an auxiliary variable that implicitly defines a prior over the HR outcome (cf. Figure 1). The model is trained by aggregating the predicted HR outcomes and enforcing consistency with the observed low-resolution (LR) measurements. This enables causal effect estimation, counterfactual reasoning, and the computation of HR estimates from LR inputs.

Because no existing method targets this estimand, our contribution is the problem formulation first and the algorithm second, and we prioritize making the assumptions and limits of the setting explicit over benchmarking against methods built for a different task.

2 Problem Setup

For notational simplicity and without loss of generality, we assume that all regions share the same subregion structure, interventions are binary, and all variables are scalar rather than vector-valued.

We consider N regions, each divided into M subregions. The LR *treatment data* is a vector $\hat{T} \in \{0, 1\}^N$, where each entry $\hat{t}_i$ indicates whether an intervention occurred in region i . We assume that, when present, the intervention affects all subregions within that region.

The LR *outcome data* is a vector $\hat{Y} \in \mathbb{R}^N$, obtained by applying a row-wise aggregation function (e.g., summation or averaging) to an unobserved HR outcome matrix $Y \in \mathbb{R}^{N \times M}$, where $y_{i,j}$ denotes the outcome value for subregion j within region i . The HR *contextual data* (covariates) is given by a matrix $C \in \mathbb{R}^{N \times M}$, where $c_{i,j}$ denotes the contextual value for subregion j in region i .

Finally, we assume that N and M are perfect squares, allowing us to represent the spatial structure as a $\sqrt{N} \times \sqrt{N}$ grid of regions, each containing a $\sqrt{M} \times \sqrt{M}$ grid of subregions (see Figure 1 for an example with $N = M = 16$ and $\sqrt{N} = \sqrt{M} = 4$).

Goal. Our goal is to learn a function that estimates local causal effects and to disaggregate coarse outcomes into fine-grained predictions. We formalize this objective in the next section.

Generalizations. The setup admits several natural extensions. Variables (such as c_{ij}) can be vector-valued, treatments can be continuous instead of binary, and the observed LR treatment $\hat{T}$ can be interpreted as an aggregation of an underlying HR treatment matrix that varies within a region. The framework also extends to spatio-temporal settings, where T , Y , and C evolve over time. More generally, additional structure such as hidden confounding, mediation, or non-trivial treatment assignment can be incorporated; we discuss these cases in Section 4. Naturally, the spatial layout need not follow a regular grid, which we adopt only for illustration.

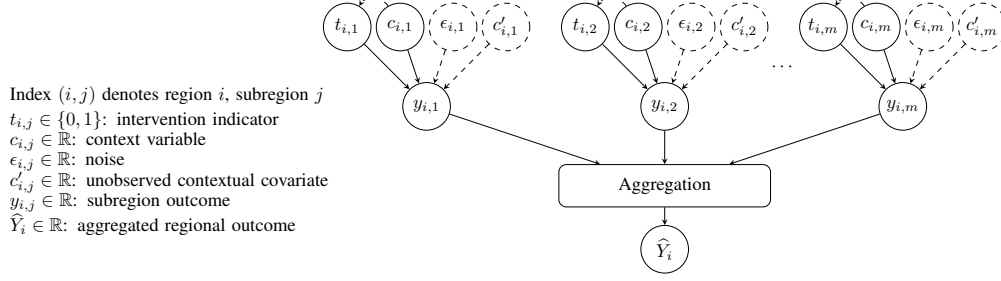


Figure 2: Causal graph for one region. All functional relations are shared among subregions and regions. Variables could also be vector-valued. Unobserved covariates and a causal link from context to treatment are optional. The aggregation is typically a mean or a sum.

Causal Assumptions. Depending on the experimental setting, different assumptions may hold or be relaxed. We may assume a known causal graph, which constrains the model and reduces the risk of spurious explanations, particularly important when contextual variables are high-dimensional and correlated, but only a subset is causally relevant. We typically assume no hidden confounding, meaning all relevant variables affecting the outcome are observed, allowing interventions to be interpreted causally up to reasonable noise. Causal effect identification also requires variation in interventions across regions. If the aggregation function from subregional to regional outcomes is known (e.g., a mean or sum), disaggregation becomes more tractable. Moreover, we typically assume that the treatment does not depend on the context values.

3 Our Method: CLAM

Our method combines two insights. First, we represent the causal disaggregation problem using a structural causal model formalism. This formalism encodes assumptions and specifies the missing pieces (functions, parameters). Second, we train an ML model to infer the missing pieces where we use the LR data as a loss signal to estimate the most likely HR data and functional relationships.

3.1 Structural Causal Model

Each region–subregion pair (i, j) is associated with a treatment $t_{i,j}$, context $c_{i,j}$, noise $\epsilon_{i,j}$, and an optional hidden confounder (or non-confounding covariate) $c'_{i,j}$ (cf. Figure 2). A shared structural function maps these inputs to a subregional outcome $y_{i,j}$, and then aggregates the subregional outcomes to a regional outcome $\hat{Y}_i$:

$$y_{i,j} = f_{\theta}(t_{i,j}, c_{i,j}, \epsilon_{i,j}, c'_{i,j}), \quad \hat{Y}_i = g_{\phi}(\{y_{i,j}\}_{j=1}^M). \quad (1)$$

Each subregion is modeled by a local functional mechanism, with uncertainty captured via $\epsilon_{i,j}$. Although the functions $f_{\theta}(\cdot)$ and $g_{\phi}(\cdot)$ are shared across regions, region-specific dynamics can still be modeled through context variables (e.g., region indicators or positional encodings). The difference between $c'_{i,j}$ and $\epsilon_{i,j}$ is that $\epsilon_{i,j}$ is i.i.d. noise, while $c'_{i,j}$ is structured, for example spatially autocorrelated or shared across observations in time.

In general, $g_{\phi}(\cdot)$ can be any function parameterized by ϕ . For simplicity, we typically assume the aggregation is a sum, the noise is additive, and there are no hidden confounders. Then the model simplifies to:

$$\hat{Y}_i = \sum_{j=1}^M (f_{\theta}(t_{i,j}, c_{i,j}) + \epsilon_{i,j}) = \sum_{j=1}^M f_{\theta}(t_{i,j}, c_{i,j}) + \sum_{j=1}^M \epsilon_{i,j}. \quad (2)$$

If we want to simplify further, we can replace the flexible estimator $f_{\theta}(\cdot)$ with a linear one, where $\theta = (\alpha_1, \beta_1, \alpha_2, \beta_2)$. Conceptually, we use two linear models, one for treated regions and one for untreated regions, each relating the local context $c_{i,j}$ to the outcome:

$$\hat{Y}_i = \sum_{j=1}^M (t_{i,j} (\alpha_1 + \beta_1 c_{i,j}) + (1 - t_{i,j}) (\alpha_2 + \beta_2 c_{i,j})) + \sum_{j=1}^M \epsilon_{i,j}. \quad (3)$$

3.2 Training

The model is trained by minimizing the discrepancy between the observed LR outcomes $\hat{Y}$ and the predicted aggregated outcomes μ , typically using an MSE loss: $\mathcal{L}_{\theta, \phi} = \frac{1}{N} \sum_{i=1}^N (\mu_i - \hat{Y}_i)^2$. Note that we use $\hat{Y}_i$ for the observed regional outcome and $\mu_i = g_{\phi}(\{f_{\theta}(t_{i,j}, c_{i,j})\}_{j=1}^M)$ for the corresponding prediction, which under zero-mean additive noise is the conditional expectation of the aggregate.

Importantly, the loss is only observed at the aggregated (region) level. The model therefore does not receive direct supervision for individual subregions; instead, it must learn a subregional mechanism that, when combined, is consistent with the observed aggregated outcomes.

When relaxing the assumptions from Section 2, training becomes more complicated. For example, if the aggregation function, $g_{\phi}(\cdot)$, is unknown, it must be learned jointly with the local mechanism $f_{\theta}(\cdot)$, which leads to a coupled estimation problem. In general, $g_{\phi}(\cdot)$ can be implemented using an (order-invariant) set neural network (e.g., [10]). This is conceptually similar to settings in graph neural networks, where both the message-passing mechanism and the aggregation function are learned [11].

Similarly, if treatments are not uniformly applied across subregions, the underlying HR treatment matrix must be inferred jointly during training. More generally, additional latent variables (e.g., unobserved confounders) can be incorporated and estimated within the same framework. This includes uneven delivery, for example a vaccination campaign reaching subregions to differing degrees.

3.3 Downstream Applications

Training yields the predictive mechanism $f_{\theta}(\cdot)$, from which several causal quantities can be derived. The most direct is what we call the *local conditional average treatment effect* (LOCATE), a version of the conditional average treatment effect (CATE) conditioned on subregion-specific factors:

$$E_{i,j} = f_{\theta}(1, c_{i,j}) - f_{\theta}(0, c_{i,j}), \quad (4)$$

i.e., the treatment–control difference at the subregional level. Aggregating these quantities yields ATEs for regions, populations, or environments; with continuous treatments, the same mechanism gives conditional dose–response curves. Note that LOCATE is noise-marginalized.

Beyond effect estimation, the framework supports outcome *disaggregation*—mapping coarse regional outcomes back to fine-grained predictions—and counterfactual reasoning, e.g., predicting outcomes under hypothetical intervention placements or alternative contexts. In the simplest case, this amounts to evaluating $f_{\theta}(\cdot)$ under new inputs. For counterfactuals, when the noise term is believed to carry realization-specific information, inferred noise values can be reused to produce sharper, sample-specific counterfactuals rather than population-level averages.

The framework can also recover latent HR treatment locations by representing them as trainable variables (cf. Exp. 2 in Appendix D.2). With time-series data, CLAM can further infer latent variables that modulate causal effects, provided temporal variation supplies enough constraints (cf. Exp. 3 in Appendix D.3).

3.4 Identifiability

We do not provide hard identifiability results, and a complete treatment is left as an open problem. Appendix A states what is obvious and what is known, and gives directions for studying the question properly. In short, recovery of the mechanism and recovery of the high-resolution outcome map have to be distinguished, and simple counterexamples show that neither is pinned down without further assumptions. Without noise and without the need to interpolate, the problem reduces to a linear equation system, which makes the required variability in the data explicit. Beyond that, the closest existing result is the consistency analysis of Zhang et al. [12], which applies to an idealized version of our setting. What the optimizer actually finds, and how the model behaves under interpolation and extrapolation, remain open.

3.5 Relationship to Causal Deabstraction

We view CLAM as an instance of *causal deabstraction*: recovering a fine-grained causal mechanism from coarse, aggregated observations. While prior work studies when causal structure is preserved under abstraction [9, 13], we consider the inverse direction: inferring a high-resolution model that is consistent with observed aggregates. Appendix B formalizes this view and clarifies the causal semantics. Aggregation deletes the pairing between outcomes and subregions. HR covariates and an invariant causal mechanism [14, 15] can partially restore it when covariate compositions vary across regions.

4 Exploiting Context Across Motifs in the Causal Graph

Throughout the paper, we focus on the simplest causal motif: a treatment T acting on an outcome Y , with a contextual covariate C that modulates the effect. Real-world settings can exhibit a much richer variety of causal structures, and the role of high-resolution context, C , shifts with the motif: sometimes it must be conditioned on, sometimes ignored, sometimes leveraged for new identification strategies. Do-calculus tells us *whether* an effect is identifiable given the graph; this section asks *how* the LR/HR distinction interacts with that question. Here, we provide an outlook in which we sketch how HR context can be exploited in common causal graph motifs.

Baseline: $C \rightarrow Y, T \rightarrow Y$. The default setting throughout this paper is the simplest motif: T and C are independent inputs, both pointing into Y , with no edge between them. C is not a confounder, only a covariate that modulates the local effect of T , and the backdoor criterion holds trivially and gives rise to the LOCATE in Eq. (4). The role of the HR resolution is purely informational: C provides multiple “views” of the same regional outcome and thereby constrains the shared mechanism $f_{\theta}(\cdot)$.

Confounder: $C \rightarrow T, C \rightarrow Y, T \rightarrow Y$. When the context C also drives whether (or by how much) a region is treated, it becomes a confounder. The backdoor criterion then requires conditioning on C . Treating T as exogenous yields biased estimates; modeling the treatment-assignment mechanism at HR is what makes the adjustment correct. Exp. 5 (Appendix F) instantiates this motif and shows that ignoring the dependence inflates the MSE on local effects. A concrete real-world example to illustrate this problem is a wildfire-prevention program: forested counties with dry summers are more likely to receive funding, and those same counties also have higher baseline fire incidence, so the program’s effect is conflated with the underlying risk.

Collider: $T \rightarrow C, Y \rightarrow C$. A collider is a variable influenced by both the treatment and the outcome. Conditioning on such a variable opens a spurious path between T and Y , so naïvely passing C as an input to $f_{\theta}(t, c)$ would bias the estimated effect. Nevertheless, HR collider information may still be useful: since C is partly determined by the HR data $y_{i,j}$, it can provide fine-grained information for the reconstruction of Y or serve as a diagnostic check. For example, consider a regional vaccination campaign (T), where the outcome of interest is infection rates (Y), and the HR variable C is neighborhood-level pharmacy purchases of rapid tests. The campaign may increase public attention and test purchases through its messaging, while actual local infections also increase demand for rapid tests.

A possible way to exploit this structure is to invert the usual roles: instead of using C as a covariate of the outcome, we treat C itself as the HR target and learn a local mechanism $f^{\text{inv}}(t, y) = c$ that maps treatment and outcome to the collider. Under this setup, CLAM’s aggregation-consistency principle still applies, but it is used to jointly learn the mechanism and the disaggregation of T and Y against the observed HR values of C . We leave a concrete implementation to future work.

Mediator: $T \rightarrow C, C \rightarrow Y, T \rightarrow Y$. A mediator C lies on the causal pathway from T to Y , so part of the treatment effect is transmitted through C rather than acting directly. Naïve estimation that conditions on C cannot separate the direct effect of T from the indirect effect through C . When the high-resolution mediator is observed, one can instead jointly learn the treatment-to-mediator, mediator-to-outcome, and direct treatment-to-outcome mechanisms. Only the two outcome-side mechanisms rely on the aggregated regional observations, since the treatment-to-mediator mechanism is supervised directly by the observed $c_{i,j}$. This enables a decomposition of the learned effect into a

direct component $T \rightarrow Y$ and an indirect component $T \rightarrow C \rightarrow Y$. A concrete example is a regional advertising campaign observed at low resolution, whose effect on regional sales, also observed at low resolution, is partly direct and partly mediated by word-of-mouth dynamics reflected in local online discourse observed at high resolution.

Spill-over. A subregional outcome may depend on neighboring treatments and contexts: interventions can spill over across space. The local mechanism is then no longer purely local, and aggregated observations alone cannot distinguish a strong local effect from a weaker effect that diffuses outward. HR context can help by exposing spatial structure; technically, one can replace the pointwise $f_{\theta}(\cdot)$ with a grid-based model. In principle, the model can learn both direct local effects and spill-over patterns from aggregated outcomes. A concrete example is a driving ban in one district that displaces traffic into adjacent ones, so regional air quality depends on both the ban location and surrounding road and vegetation patterns.

Time-Varying Confounding. In spatiotemporal settings, confounders may evolve over time and be shaped by past treatments. Present context can then no longer be treated as a fixed pre-treatment covariate, because it may already encode earlier interventions. This makes estimation harder: naïve adjustment can block part of the causal pathway or introduce bias [16]. CLAM can be extended by indexing $f_{\theta}(\cdot)$ over time and by explicitly modeling how past treatments, outcomes, and HR covariates affect the current state. Exp. 3 already uses temporal variation to recover a stable latent spatial factor; a natural next experiment would make this factor dynamic and treatment-dependent. For example, neighborhoods with high prior infection rates may receive additional public-health resources, which then change later vulnerability and influence future allocation decisions.

Summary. Across all motifs, the role of HR data can be summarized as follows: it provides *variation within regions* that translates into observable differences at the aggregated level. This variation can be exploited to constrain and identify fine-grained causal mechanisms, but only when the causal graph permits it.

More concretely: (i) for effect modifiers and confounders, HR structure is highly informative and can be directly leveraged for identification and adjustment; (ii) for mediators, it remains informative but requires explicitly modeling the underlying causal pathways; (iii) for colliders, it can be actively misleading if the framework is not changed accordingly.

CLAM does not replace causal identification via do-calculus: the causal graph determines whether a target effect is identifiable in principle, while CLAM provides a mechanism for estimating the corresponding fine-grained structure from aggregated data.

However, identifiability in principle does not guarantee successful recovery in practice. Learning the underlying mechanism further requires sufficient variability in the HR covariates, a well-behaved optimization problem, and effective training of the chosen function class. We discuss these limitations in more detail in Section 7.

5 Related Work

CLAM sits at the intersection of multiple research strands; an extended discussion is provided in Appendix C.

Spatial Disaggregation and Downscaling. A long line of work in statistical downscaling and spatial disaggregation infers high-resolution estimates from coarse data using interpolation, regression, or learning-based techniques [1, 2, 17, 18]. These methods are predominantly associational; recent extensions integrate causally relevant predictors [19, 20] but still rely on relatively fine-grained information and do not target interventional or counterfactual queries.

Causal Inference under Aggregation and Abstraction. Reasoning causally with aggregated data is well known to be hazardous: classical results on the ecological fallacy show that aggregate statistics may obscure or invert relationships at the unit level [21–24]. Work on causal abstraction formalizes when mappings between models at different granularities preserve interventional or counterfactual semantics [9, 25, 13], and spatiotemporal causal inference [26–32] addresses related challenges with

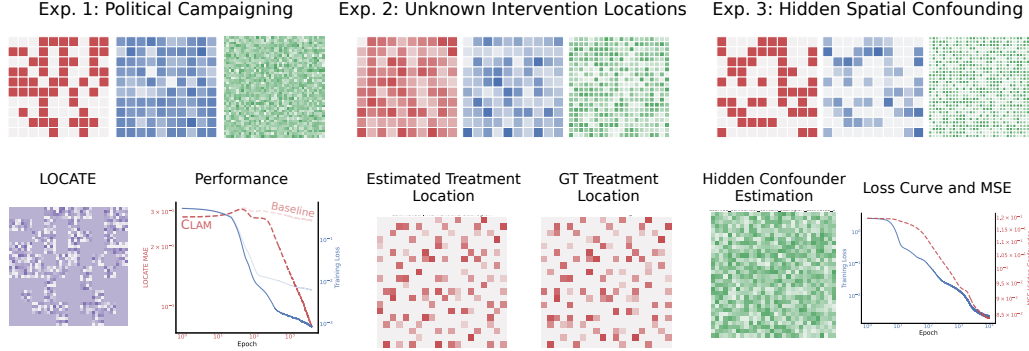


Figure 3: **Results Overview.** **Top:** Input of LR intervention locations (red), LR outcomes (blue), and HR context (green). **Bottom:** Selected inference results.

different data regimes. None of these directly target the joint problem of disaggregating outcomes *and* estimating local causal effects from coarse interventions.

Invariant Mechanisms, GNNs, and Hierarchical Models. Conceptually, CLAM relies on the principle of *invariant causal mechanisms* [14, 15, 33, 34]: the local mechanism $f_{\theta}(\cdot)$ is shared across subregions while contextual covariates drive heterogeneity. This turns aggregation into a constraint and connects our architecture to message-passing GNNs [11] and Deep Sets [10], as well as to hierarchical Bayesian models that pool information across groups [35, 36]. It also places CLAM in the setting of learning from aggregate observations [12], whose training loss coincides with ours under mean aggregation (cf. Appendix A).

6 Experiments

We evaluate CLAM on three synthetic studies and one real-world study. In the synthetic studies the ground truth mechanism is available isolate distinct capabilities of the framework under controlled conditions: heterogeneous local-effect recovery, latent intervention localization, and the reconstruction of unobserved covariates. The real-world study then probes the method on U.S. gun-violence data, where we assume that only coarse regional outcomes are observed.

Full experimental details, per-experiment figures, and additional analyses are provided in Appendices D and E, with the ablation study included in Appendix D. Two additional studies—covering confounded treatment allocation and an unknown aggregation function—are deferred to Appendix F. Our focus is qualitative: we aim to characterize the behavior of the method. Figure 3 provides an overview of the inputs and headline results. Code that runs off-the-shelf on Google Colab is provided.

6.1 Synthetic Studies

Exp. 1: Political Campaigning. We first ask whether CLAM can recover heterogeneous local causal effects from aggregated outcomes when the effect of treatment depends on subregional context. We model a binary regional intervention (campaign spending) whose impact varies with subregional wealth (z-scored income values). Regional-level covariates largely remain uninformative by construction. The reported outcome is the candidate’s relative improvement, observed only at the regional level. We compare against a naïve **baseline** (Figure 3, with confidence intervals in the Appendix) that applies a uniform disaggregation mechanism to the regional outcome and then learns the causal mechanism.

Uniform, bicubic, and ecological-regression [37] baselines all reach the same LOCATE error floor. Without HR context a method can only redistribute the regional total, so at best it assigns every subregion the regional average effect, and the residual error is the within-region spread of the true effect. CLAM operates below this floor because the shared mechanism ties each outcome to the context of its subregion. This is the gain hypothesized in the introduction, where treating disaggregation and causal inference separately caps what is achievable.

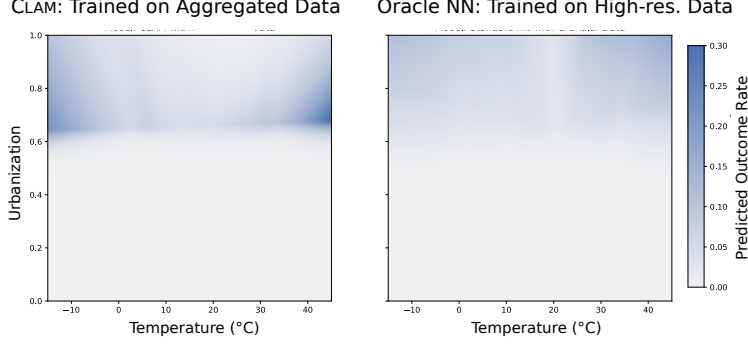


Figure 4: Side-by-side comparison of the learned rate functions $f_{\theta}(\text{temp}, \text{urban})$ for the disaggregation from CLAM (left) and the oracle (a NN with the ground truth subregional data; right). Predictions are generated across a temperature range of $[-15, 45]^{\circ}\text{C}$ and an urbanization scale of $[0, 1]$.

We use a simple neural network to estimate the causal mechanism. CLAM accurately recovers the underlying subregional causal effects, with the MAE to the ground-truth LOCATE matrix (cf. Eq. (4)) decreasing alongside the training loss. The learned model supports coherent counterfactual reasoning at both regional and subregional resolutions and correctly identifies regions whose outcome differences arise purely from contextual heterogeneity rather than mean effects. Full setup and per-subregion visualizations are given in Appendix D.1 (Figure 5).

Exp. 2: Public School Funding. Next, we test whether CLAM can recover *latent intervention locations* jointly with local causal effects when only aggregated intervention totals and outcomes are observed. Each of the 30×30 regions consists of 4×4 subregions and receives binary funding in either no subregion or exactly one unobserved subregion. Treatment effects vary with subregional socioeconomic status, and the outcome is education-score improvement. We implement $f_{\theta}(\cdot)$ using either a parametric function or a small neural network. In both cases, CLAM jointly identifies the treated subregions and accurately recovers the underlying causal-effect function. To sample a treated subregion, we use the common Gumbel-Max trick. Details are provided in Appendix D.2 and Figure 7.

Exp. 3: Spatiotemporal Effects of Heat Waves. Finally, we ask whether CLAM can reconstruct an *unobserved spatial effect modifier* from aggregated outcomes in a spatiotemporal setting. Heat waves negatively affect school performance, with heterogeneous impact depending on parental education (observed) and vegetation coverage (unobserved), the latter mitigating heat exposure. Only region-level outcomes are observed over time. Under a correctly specified linear mechanism, both the regional prediction loss and the MSE converge rapidly, and the estimated disaggregation of the vegetation is visually indistinguishable from the ground-truth vegetation map. With a flexible MLP, CLAM still recovers meaningful spatial structure but with substantially higher error. This reflects an identifiability trade-off whereby expressive models can absorb transformations of the vegetation while preserving aggregated predictions. Details are provided in Appendix D.3 (Figure 8).

Takeaways. Across the three studies, CLAM performs well in the synthetic setting, particularly for estimating causal effects. The recovery of latent variables, such as intervention locations or hidden confounders, depends on the specific experiment. Overfitting may occur, when large amounts of data are available.

6.2 Semi-Synthetic Case Study With Real-World Data: Heat and Gun Violence

We evaluate CLAM on U.S. gun-violence data from the Gun Violence Archive (GVA) for 2022–2023 [38], motivated by prior evidence linking higher ambient temperatures to increased firearm violence [39]. We treat temperature as a continuous exposure. Urbanization is the high-resolution (HR) contextual covariate, and gun-incident counts are observed at weekly resolution. A core difficulty in evaluating disaggregation methods on real data is the absence of ground truth at the fine scale. We work around this by starting from data that is natively HR (incident counts at the cell

level) and *artificially* aggregating it into LR regional counts, which serve as the inputs to CLAM. To construct a high-quality reference **baseline**, the original HR counts are only used for fitting an “oracle” that we use for evaluation.

We use $N = 100$ regions and $M = 100$ cells per region, indexed by region i , subregion j , and week w . Let $Y_{i,w}^{\text{LR}}$ be the aggregated regional count in region i during week $w \leq 52$, $c_{i,j}$ the HR urbanization covariate, and $\tilde{t}_{i,w}$ the standardized regional temperature. We model subregion-level rates as $\lambda_{i,j,w} = f_{\theta}(\tilde{t}_{i,w}, c_{i,j})$ and aggregate them as $\mu_{i,w} = \sum_{j=1}^M \lambda_{i,j,w}$, with $Y_{i,w}^{\text{LR}} \sim \text{Poisson}(\mu_{i,w})$. Training minimizes the regional Poisson negative log-likelihood. Figure 4 shows that CLAM recovers a qualitatively similar temperature–urbanization interaction to the oracle. The spatial comparison in Appendix Figure 9 shows the same pattern: plausible structure, but inflated and more diffuse local intensities, reflecting an aggregation-induced identifiability gap.

7 Limitations

CLAM relies on assumptions that are necessary given the nature of inferring fine-grained causal effects from aggregated data, but are nonetheless restrictive. We discuss the main limitations here.

Prerequisite: Do-Calculus Identifiability. CLAM is not a substitute for causal identification. The method should only be applied in settings where do-calculus (applied to a known or expert-specified causal graph) already certifies that the causal mechanism is in principle recoverable. If the causal graph is unknown or the required identification conditions fail (e.g., the backdoor criterion is violated by unmodeled confounders), CLAM will fit a model to the data that will not carry a causal interpretation.

No Identification Guarantee from Optimization Alone. Even when do-calculus certifies identifiability in principle, we do not provide a formal guarantee that our optimization procedure recovers the true causal effect. The training objective is non-convex, and the best we can reasonably claim is that, under good initialization and sufficient covariate diversity, gradient descent finds a local minimum whose value is close to the true causal effect. It is easy to come up with pathological cases in which different fine-grained disaggregations are equally consistent with the observed regional outcomes. How to detect or avoid them in practice, remains an open question.

Assumptions and Their Scope. Recovery of the local mechanism further requires (i) an invariant causal mechanism across subregions, (ii) sufficient diversity in subregional covariate distributions across regions, (iii) no hidden confounding, or its explicit modeling, and (iv) a known or learnable aggregation function.

The invariance assumption (i) is less restrictive than it may appear: although $f_{\theta}(\cdot)$ is shared across regions and subregions, region-specific dynamics can still be captured through contextual covariates such as region indicators or positional encodings. Apparent heterogeneity across regions is absorbed by the context, yielding a *conditionally* invariant model. The stronger requirement is that the contextual covariates sufficiently capture all sources of heterogeneity that are correlated with the treatment. Unobserved factors that are uncorrelated with the treatment effectively act as additional noise: they make training harder but do not bias the estimated effect. Unobserved confounders, by contrast, will generally lead to incorrect estimates unless they are explicitly modeled. The aggregation function (iv) need not be perfectly known in advance: as demonstrated in Exp. 4 (Appendix F), it can be jointly learned alongside the local mechanism provided one knows that the aggregation itself is uncertain and parameterizes it accordingly.

Identifiability Gap and Latent Variable Recovery. Even when all assumptions hold, aggregation induces an intrinsic identifiability gap: matching aggregated predictions to aggregated observations may admit multiple equally valid solutions. Our real-world study illustrates this, with CLAM systematically inflating local effect magnitudes relative to the cell-supervised oracle. The problem is amplified when latent variables are jointly inferred (e.g., intervention locations in Exp. 2, unobserved covariates in Exp. 3), since different inferred values can each correspond to a mechanism that explains the data equally well. We discuss this further in Appendix A and regard a rigorous treatment, together with practical diagnostics, as the most important direction for future work.

Limited Real-World Validation. Our evaluation is primarily qualitative and relies substantially on synthetic studies, where the data-generating assumptions are known and can be matched to the model. The real-world study uses a subregion-supervised oracle as a reference rather than a true causal benchmark. However, the causal interpretation of temperature on gun-violence intensity is supported by prior literature [39]. Robustness to realistic noise, measurement error, nonstationarity, and treatment–context dependence is explored in the additional experiments, but a more systematic benchmarking remains an important direction for future work.

8 Conclusions and Future Work

The primary contribution of this work is not a single algorithm but the articulation of a *problem class*: estimating high-resolution causal effects from low-resolution interventions and outcomes by exploiting high-resolution context. We frame this as *causal deabstraction*, inverting the standard notion of causal abstraction and recasting spatial aggregation as a constraint satisfaction problem, thereby bridging structural causal models and spatial statistics. In many real-world settings such as public health, environmental policy, and education, spatial disaggregation and causal estimation are inseparable, since interventions are implemented at coarse scales but their effects manifest locally. Treating them jointly within a structural causal model makes assumptions explicit, clarifies what is and is not identifiable, and turns causal inference into a tractable optimization problem.

CLAM is one concrete instantiation of this framework. It explicitly adjusts for subregional treatment-context confounding using only coarse regional supervision, a challenging problem in ecological inference, and jointly learns latent high-resolution variables such as exact intervention locations and spatial confounding alongside the causal mechanism. Across synthetic and real-world studies, we recover meaningful spatial and functional structure from aggregated data, while exposing the inherent limits of the setting: relative structure and causal dependencies are recoverable, but absolute local magnitudes may be systematically overestimated without fine-grained outcome supervision. We conjecture that this reflects the problem class rather than CLAM in particular.

Because the problem class is, to our knowledge, new in this explicit form, principled baselines are scarce and a complete identification theory does not yet exist. We see both as consequences of stating a new problem, and hope this paper motivates methods that supersede ours.

Although this paper primarily proposes a research framework, misuse in high-stakes domains such as epidemiology could cause harm if poorly identified local effects guided interventions or resource allocation; outputs should be accompanied by uncertainty estimates, sensitivity analyses, and domain expertise. Disaggregation itself may also raise privacy concerns, since inferred high-resolution patterns can reveal sensitive local information.

Several directions stand out. Theoretically, the most pressing need is a rigorous identifiability analysis: which combinations of aggregation function, contextual diversity, and structural restrictions guarantee recovery of the local mechanism, and which only up to monotone or scale transformations? Methodologically, principled regularization, hierarchical priors, or Bayesian formulations could control aggregation-induced bias and provide uncertainty quantification. Relaxing the shared-mechanism assumption, integrating causal discovery over contextual variables, incorporating instrumental-variable settings, and extending to spatiotemporal aggregation are natural next steps. We hope that framing CLAM as a first concrete entry in a broader problem class makes these extensions easier to pursue.

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A Identifiability

One of the key questions in causal modeling is identifiability. We note upfront that a complete identification theory is deliberately left as an open problem. It spans recoverability conditions, non-convex optimization, and the role of the loss and the noise model. Part of our contribution is to state the problem precisely enough that the remaining questions can be attacked separately. The discussion should be carried out for all three versions given in Section 3.1, where identification becomes simpler as the model class is restricted. Here we only give some directions and state the obvious, as a basis for further research.

Identifiability of What. There are two separate objects to consider: (i) recovery of the mechanism f_θ and (ii) recovery of the high-resolution outcome map $\{y_{i,j}\}$. The two are related but not equivalent. Knowing the mechanism does not by itself pin down the high-resolution outcome map (see the counterexample below). Conversely, recovering the high-resolution outcome map determines the mechanism at the observed input configurations $(t_{i,j}, c_{i,j})$, though not a symbolic description of it, and not its behavior off the observed support. One could in principle also ask about identification of the noise, but this matters only for counterfactual queries or under a complicated noise model, neither of which is the focus of this work.

Counterexample. Consider a case similar to Equation (2), but without noise and assuming that no region receives treatment. Take two regions A and B , each with two subregions, with context values $(0, 1)$ and $(0.5, 0.5)$ respectively. Both regional aggregates equal 1. There is an infinite family of $(f_\theta(\cdot), g_\phi(\cdot))$ pairs consistent with this observation. For instance, the identity map for $f_\theta(\cdot)$ (i.e., $f_\theta(\cdot, c_{i,j}) = c_{i,j}$) together with a sum aggregate, but equally the constant map $f_\theta(\cdot) \equiv 1$ together with a mean aggregate. Neither the mechanism nor the subregional outcomes are pinned down. This also holds when $g_\phi(\cdot)$ is known to be, for instance, a sum operation. Then $f_\theta(\cdot)$ could still be the identity or just give 0.5 for each input. Even if we knew the mechanism (e.g., $f_\theta(\cdot)$ is the identity and $g_\phi(\cdot)$ is a sum), this would not help us identify the subregional outcome map of region A , since an aggregate of 1 under a sum is equally consistent with the subregional outcomes being $(0, 1)$, or $(0.5, 0.5)$, or any other pair summing to one.

Linear Equation System. Under the assumption of no noise, no required interpolation, and a sum aggregation, the problem reduces to a linear system. We treat each distinct tuple $(t_{i,j}, c_{i,j})$ as one entity and treat it as an unknown variable (e.g., a represents the exact tuple $(t_{A,1}, c_{A,1})$), whose value is the corresponding subregional outcome. Each region contributes one equation. Identification then follows directly from the solvability of that system. The example above yields $a + b = 1$ and $2d = 1$, where a and b are the subregions of the first region and d is the (repeated) subregion value of the second. The second region is identified, that is, $y_{B,j} = 0.5$ for all subregions j of B . The first is not. Typically, however, we do not face the underspecified regime illustrated above but an overspecified one, in which we seek the solution with the smallest error.

Aggregate ML. CLAM’s training loss is closely related to the one studied by Zhang et al. [12]. In the special case of mean aggregation with additive Gaussian noise, and when the number of subregions is fixed, their loss is exactly the aggregate MSE, which matches Equation (2) up to a constant. Their Proposition 7 then shows that any parameter explaining the aggregates equally well

gives the same mechanism almost everywhere. Their proof uses independence of the covariates within a region, to make the cross terms in the population loss nonnegative. What is not covered is spatial autocorrelation between subregional contexts. We still find their notion of consistency up to an equivalence relation to be the right language here to study this problem.

Non-Linear Aggregation. A nonlinear aggregation adds a further complication. The observed regional outcome is $g_\phi(\cdot)$ applied to the realized subregional outcomes, whereas the model applies $g_\phi(\cdot)$ to their expectations $f_\theta(t_{i,j}, c_{i,j})$. For a sum the two coincide and the objective is exact. For a nonlinear $g_\phi(\cdot)$ they do not. This is a question of identification as the true mechanism is no longer the minimizer of the loss. Sampling the noise in the forward pass closes the gap, at the price of a known noise distribution and noisier gradients.

Open Problems. Several questions remain open. First, even when the true high-resolution outcome and mechanism are, under suitable interpolation assumptions, the ones minimizing the loss, it is unclear under which conditions the solver actually finds them; this is particularly delicate in double-ML approaches. More importantly, we regard the issues arising under real-world conditions as the larger problem: what happens under interpolation and extrapolation beyond the observed support, which is tied directly to the inductive bias of the network; what happens when the noise model is more complicated than additive; and what variability in the data is precisely required for recovery.

High-Level Conclusion. There is a trade-off between realism and identification guarantees. Restricting the mechanism until identification can be argued cleanly buys guarantees about a model that may no longer describe the process of interest; admitting a flexible mechanism keeps the model faithful but forfeits those guarantees. For practical purposes we tend to use the more expressive neural mechanism and rely on the empirical observation that SGD recovers a better mechanism than the linear alternative, even though the linear case is the one we can analyze. We view this as a deliberate choice rather than a resolved question, and the diagnostics below are what we substitute for the guarantee.

As piratical guidance, we recommend using bootstrapping when unsure and, where available, external validation against higher-resolution ground truth. The practitioner may also fit several model classes and check whether the reconstructed high-resolution variable converges across them. Agreement is not proof of identification, but disagreement is reliable evidence of its absence.

B Causal Deabstraction Framework

In this section, we situate our work within the broader causal inference literature. Specifically, we frame this as a novel instance of *causal deabstraction* [9, 13]. Our methodology is informed by the framework of SCMs and is guided by the principle of ICM.

B.1 Structural Causal Framework

The structural causal model introduced in Section 3.1 can be formalized in generative form as follows. Each spatial unit (region) $i \in \{1, \dots, N\}$ is composed of M subregions indexed by $j \in \{1, \dots, M\}$. For each subregion (i, j) , we posit the following SCM:

$$\begin{aligned} c_{i,j} & \text{ observed (exogenous)} \\ \epsilon_{i,j} & \sim P_\epsilon \\ t_{i,j} & = T_i \in \{0, 1\} \\ y_{i,j} & = f_\theta(t_{i,j}, c_{i,j}, \epsilon_{i,j}) \\ \hat{Y}_i & = g_\phi(\{y_{i,j}\}_{j=1}^M) \end{aligned}$$

Here, $c_{i,j}$ denotes a high-resolution contextual covariate for subregion j within region i , which is observed and exogenous, meaning it is not caused by T or Y , and $\epsilon_{i,j}$ represents exogenous noise drawn from P_ϵ . The binary intervention $t_{i,j} = T_i \in \{0, 1\}$ is assigned at the regional level and shared across all subregions. The function $f_\theta(\cdot)$ encodes the local causal mechanism, shared across all subregions, that maps the intervention, context, and noise to the subregional outcome $y_{i,j}$. The

aggregation function $g_\phi(\cdot)$ maps the collection of subregional outcomes $\{y_{i,j}\}_{j=1}^M$ to the regional outcome $\hat{Y}_i$.

In the main section, we considered simplified cases, such as sum aggregation, additive noise, and the absence of hidden confounders, to provide intuition. The formulation here generalizes that view by expressing the structural causal model in explicit stochastic form, specifying how interventions, contextual variables, and noise are generated, while making clear that both $f_\theta(\cdot)$ and $g_\phi(\cdot)$ are shared across regions.

B.2 Invariant Causal Mechanism Assumption

A central assumption in CLAM is that the causal function $f_\theta(\cdot)$ is *invariant* across all subregions and regions. That is, while the covariates $\{c_{i,j}\}$ may vary across regions due to differences in local composition, the functional form $f_\theta(t, c)$ does not change. This assumption reflects the principle of *invariant causal mechanisms* [15, 14, 33], which posits that causal relations are stable across environments unless directly intervened upon.

We treat each region i as an *environment* defined by its observed covariate composition $\{c_{i,j}\}_{j=1}^M$, but governed by a shared mechanism $f_\theta(\cdot)$. The variability in covariate composition across regions allows us to disentangle $f_\theta(\cdot)$ from the observed aggregated outcomes $\hat{Y}_i$.

B.3 Deabstraction via Invariance

We propose to view the task of estimating $f_\theta(\cdot)$ from aggregated outcomes as an instance of *causal deabstraction*, that is, inferring a latent, high-resolution causal model that is consistent with a lower-resolution aggregated model. This perspective is dual to recent work on causal abstraction [9, 13], which studies when a coarse model preserves the counterfactual semantics of a fine-grained one. Here, we reverse the direction: we seek to recover a fine-grained causal mechanism that is *compatible* with the observed coarse-level effects.

This is possible due to two key properties:

1. The aggregation function $g_\phi(\cdot)$ is consistent and applied uniformly across regions.
2. The observed covariate compositions $\{c_{i,j}\}_{j=1}^M$ differ across regions, which induces identifiable variation in the aggregates $\hat{Y}_i$ under the shared mechanism $f_\theta(\cdot)$.

Given sufficient diversity and the assumption of causal invariance, the aggregated outputs provide a supervisory signal to recover the latent causal function.

B.4 Causal Semantics via Do-Calculus

Under the assumption of no hidden confounding, the causal effect of the intervention T_i on a subregional outcome $y_{i,j}$ is identified via the backdoor criterion:

$$\mathbb{E}[y_{i,j} \mid do(T_i = t), c_{i,j}] = \int f_\theta(t, c_{i,j}, \epsilon_{i,j}) p(\epsilon_{i,j}) d\epsilon_{i,j}$$

In Exps 1–4, we assume $T_i \perp\!\!\!\perp \epsilon_{i,j} \mid c_{i,j}$, and that T_i is either randomized or independent of $C_{i,j}$. Under this setting, the expectation can be approximated via observational data. However, because only the aggregate outcome $\hat{Y}_i$ is observed, we must rely on the consistency constraint:

$$\hat{Y}_i \approx \sum_{j=1}^M f_\theta(T_i, c_{i,j})$$

The training procedure thus finds the function $f_\theta(\cdot)$ that jointly explains all observed aggregates under this constraint, while enforcing that $f_\theta(\cdot)$ is invariant across regions. In this sense, the estimation of $f_\theta(\cdot)$ becomes a constraint satisfaction problem, where causal invariance and aggregation consistency define the feasible solution space. In Exp 5, this conditional independence assumption is deliberately violated by construction, and valid estimation requires explicitly modeling the treatment allocation mechanism.

B.5 Implications

The assumption of an invariant subregional mechanism plays a central role in enabling causal inference in our disaggregated setting. By treating the coarse observations as aggregates of fine-grained causal effects and leveraging variation in covariate compositions, CLAM performs inference not merely on statistical associations but on causal mechanisms that are meaningful and robust to changes in population structure. All in all, we contribute a formal and algorithmic instantiation of *causal deabstraction* from low-resolution outcomes and high-resolution covariates.

C Extended Related Work

Our work builds on and extends three main strands of research: spatial disaggregation, causal inference under aggregation, and causal abstraction.

Statistical Downscaling and Disaggregation. Statistical downscaling methods aim to infer high-resolution estimates from coarse data, using spatial interpolation, small area estimation [40], or learning-based techniques. These have been widely applied in climate and environmental sciences [17, 41, 42], including recent work on probabilistic downscaling using Gaussian processes [18, 43]. However, such methods typically model associational patterns and do not support interventional or counterfactual reasoning.

To address this, some recent approaches integrate causal reasoning into downscaling pipelines, e.g., by identifying causally relevant predictors [19, 20]. Yet, these still rely on observable variables and assume access to relatively fine-grained information. In contrast, our approach estimates high-resolution *causal effects* using only aggregated outcome data and high-resolution covariates.

The challenges of reasoning with aggregate data are well-documented. Classical results on the ecological fallacy show that aggregated statistics may obscure or even invert causal relationships [44, 21, 45]. Earlier work on aggregation and disaggregation in optimization similarly highlights the complexity introduced when latent heterogeneity is present but unobserved [22]. CLAM explicitly addresses these challenges by modeling the aggregation process and learning disentangled causal effects at the subregional level.

Causal Deabstraction. Structurally, our approach is inspired by recent work on causal abstraction, which studies mappings between causal models at different levels of granularity [9, 25, 13]. These works formalize when such mappings preserve counterfactual or interventional semantics, but do not address learning disaggregated effects from data. Finally, our work relates to spatiotemporal causal discovery [46], but differs by operating in a setting where temporal data is limited and only aggregate observations are available.

Graph Neural Networks. Conceptually, our architecture is related to message-passing graph neural networks [11]. In a GNN, pairwise messages are computed and aggregated within a single update step. Similarly, our model learns an update function $f_\theta(\cdot)$ that operates on ordered inputs, and optionally an aggregation function $g_\phi(\cdot)$ that is permutation-invariant, akin to a Deep Sets model [10]. As in GNNs, we share the parameters of the structural function across nodes or regions.

Invariant Causal Mechanisms. Our work also relates to the principle of *invariant causal mechanisms*, which posits that the functional relationships between variables remain stable across different environments or interventions [14, 15]. This idea has been used for causal discovery [33] and domain generalisation [34], and is particularly relevant for spatial disaggregation where intervention effects may vary locally but share stable dependencies on contextual covariates. CLAM incorporates this principle by enforcing that the learned local causal mechanism $f_\theta(\cdot)$ is shared across all subregions while allowing spatial heterogeneity to emerge through contextual variables.

Hierarchical Bayesian Model. The design also parallels hierarchical Bayesian models [35, 36], where group-specific effects are drawn from a shared prior distribution and information is partially pooled across groups. In CLAM, the local mechanism $f_\theta(\cdot)$ plays the role of the shared prior, and contextual covariates capture structured variation across subregions, allowing data-sparse subregions to benefit from patterns learned in others.

Spatiotemporal Causality. Spatiotemporal causal inference has received growing attention as researchers seek to understand how interventions and their effects evolve across both space and time. A recent work presents a causal framework for studying the interplay between conflict and forest loss in Colombia, emphasizing the need to model spatial and temporal dependencies jointly [26]. Extending this perspective, methods proposed in [27] estimate heterogeneous treatment effects in settings where economic assistance moderates the impact of airstrikes on insurgent violence, while approaches introduced in [28] accommodate arbitrary spill-over and carryover effects across space and time. Network-based settings with temporal dynamics have also been explored. In [29] the authors study bipartite causal inference with interference in time series data, and in [32] methods are developed to analyse the causal effects of modified treatment policies under network interference. Recent methodological innovations include GST-UNet [30], a deep learning architecture for spatiotemporal causal inference under time-varying confounding, and consistency-guided aggregation for causal discovery in multivariate spatiotemporal data introduced in [31]. Together, these works highlight the diversity of approaches to spatiotemporal causality, spanning structural models, interference-aware estimation, and machine learning-driven inference.

Details of Exp. 1: Political Campaigning

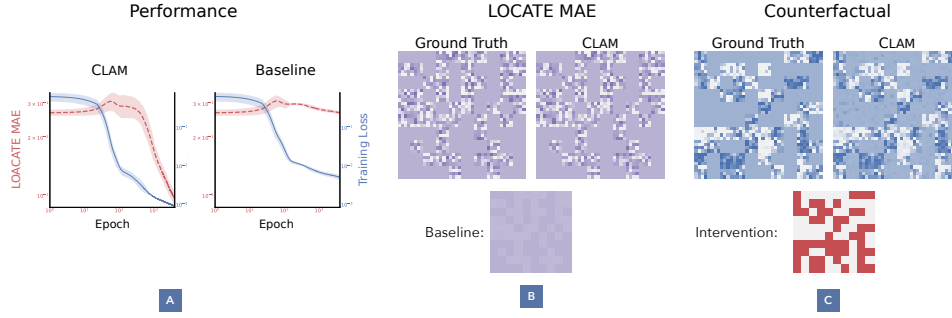


Figure 5: Evaluation of the learned outcomes, causal effect estimates, counterfactual predictions, and training behavior.

D Details of Main Experiments

D.1 Exp. 1: Political Campaigning

Context. We consider a simple political campaigning setting in which a candidate either spends campaign resources in a region or does not. The treatment is binary at the regional level and is inherited by all subregions. The outcome is the candidate’s relative improvement, but it is observed only as a regional aggregate. The effect of campaigning depends on subregional context, here interpreted as standardized wealth. Thus, two regions can have similar aggregate context and similar aggregate outcomes while still containing different subregional causal effects.

Setup. We use $N = 100$ regions arranged on a 10×10 grid. Each region contains $M = 16$ subregions arranged on a 4×4 grid. The treatment matrix is $T \in \{0, 1\}^{N \times M}$. Exactly half of the regions are treated, and within a treated region all subregions have treatment value 1; within a control region all subregions have treatment value 0.

The context matrix $C \in \mathbb{R}^{N \times M}$ contains one continuous contextual value per subregion. We sample the entries from a standard Gaussian distribution and then subtract the mean within each region, so that each row of C has mean zero. This makes the regional average context uninformative by construction. We then randomly swap entries across the matrix to introduce additional variation while preserving the overall structure. The observed outcome is generated at the subregional level and then aggregated to the region level by averaging.

The ground-truth outcome function is

$$f(t, c) = \begin{cases} 1, & t = 0, \\ \text{interp}(c), & t = 1, \end{cases}$$

where $\text{interp}(c)$ is the piecewise linear interpolation through the anchor points $\{(-2, 2), (0, 1), (1, 0)\}$.

Thus, untreated subregions have outcome 1, while treated subregions have an outcome that depends nonlinearly on the local context c . The noisy subregional outcome is

$$y_{i,j} = f(t_{i,j}, c_{i,j}) + \epsilon_{i,j}, \quad \epsilon_{i,j} \sim \mathcal{N}(0, 0.1^2),$$

and the observed regional outcome is

$$\hat{Y}_i = \frac{1}{M} \sum_{j=1}^M y_{i,j}.$$

The ground-truth local causal effect, which we use only for evaluation, is

$$E_{i,j} = f(t_{i,j}, c_{i,j}) - f(0, c_{i,j}).$$

Ablation Study of Exp. 1

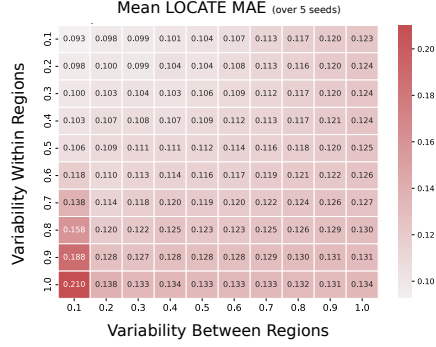


Figure 6: We evaluate the effect of varying the variance of regional context means and, for each fixed context mean, the inter-regional variance. Results are averaged over 5 random seeds for each parameter combination, and uncertainty bounds represent the corresponding variability.

Training. We train a small neural network $f_{\theta}(t, c)$ with two inputs, treatment and context, and one scalar output. The model has two hidden layers with 32 hidden units each, ReLU activations, dropout with probability 0.05, and is trained with AdamW using learning rate 10^{-3} and weight decay 10^{-3} for 4000 epochs. Importantly, CLAM is trained only from the regional aggregate signal. For each region, we average the model’s subregional predictions and minimize

$$\mathcal{L}_{\text{CLAM}} = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{M} \sum_{j=1}^M f_{\theta}(t_{i,j}, c_{i,j}) - \hat{Y}_i \right)^2.$$

Baseline. We compare against a uniform disaggregation baseline. Since the true subregional outcomes are not observed, this baseline assigns the same regional outcome to every subregion,

$$\tilde{y}_{i,j} = \hat{Y}_i.$$

It then trains the same neural network architecture using an element-wise loss,

$$\mathcal{L}_{\text{base}} = \frac{1}{NM} \sum_{i=1}^N \sum_{j=1}^M (f_{\theta}(t_{i,j}, c_{i,j}) - \tilde{y}_{i,j})^2.$$

This gives the baseline a dense subregional loss signal, but the signal is based on a uniformity assumption that removes the within-region heterogeneity needed to recover local causal effects.

Evaluation. We evaluate whether the learned model can recover the fine-grained causal structure from aggregate observations (Figure 5). We report the training loss and LOCATE MAE (cf. Eq. (4)) over training (Figure 5a). Confidence intervals (95%) are bootstrapped and based on 20 runs. As expected, both decrease for CLAM, while the baseline cannot learn the causal mechanism. We also report the full LOCATE matrix, comparing the ground truth with the one inferred by CLAM, in Figure 5b. The recovered causal effect is almost indistinguishable from the ground truth, while the baseline, as expected, cannot reconstruct the underlying disaggregation. Lastly, we construct a counterfactual treatment matrix (Figure 5c) by inverting the observed treatment assignment, $T^{\text{cf}} = 1 - T$, and compare the ground-truth counterfactual outcome $f(T^{\text{cf}}, C)$ with the neural-network estimate $f_{\theta}(T^{\text{cf}}, C)$. This tests whether the learned mechanism supports coherent counterfactual prediction beyond the observed treatment assignment.

Ablation. To test when this recovery is possible, we repeat the experiment while varying two properties of the context matrix. The first is the standard deviation of the regional context means, denoted σ_{reg} . The second is the standard deviation of context values within each region, denoted σ_{subreg} . For each pair $(\sigma_{\text{reg}}, \sigma_{\text{subreg}})$, we run the experiment over multiple random seeds and average the final LOCATE MAE. This ablation shows that high variability within regions is associated with a smaller error.

Details of Exp. 2: Unknown Intervention Location

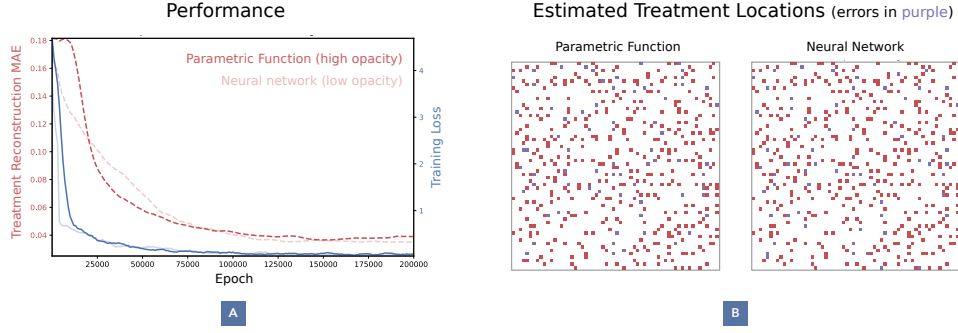


Figure 7: Performance of the training loss and MSE of the local-effect estimation, alongside ground-truth causal effects, estimated causal effects, and the corresponding estimation errors across subregions.

D.1.1 Ablation of Exp. 1: Political Campaigning

Our setting targets low-variability aggregate regimes, where regional outcomes are nearly constant and the inverse problem is underdetermined: many fine-grained causal explanations can yield indistinguishable regional observations. The key idea is that high-dimensional subregional covariates can restore information by exposing the shared causal mechanism $f_{\theta}(\cdot)$ to diverse contextual compositions within each aggregate unit. Exp. 1 and the corresponding ablation support this mechanism: as subregional heterogeneity increases, recovery of the true causal effect improves, even when aggregate outcomes remain similarly uninformative. By contrast, varying regional-level heterogeneity has a weaker effect, indicating that the gains do not come primarily from more variable aggregate outcomes, but from richer within-region contextual structure. See Figure 6 for details. Thus, causal deabstraction becomes possible in otherwise weakly identified settings when subregional covariate variation, together with an invariant causal mechanism, provides enough compositional diversity to distinguish the underlying fine-grained causal function.

D.2 Exp. 2: Public School Funding vs Improvement in Outcomes

Context. We examine how public spending on schools affects educational outcomes when the exact locations of the expenditures are *unknown*. Each administrative region contains multiple subregions, interpreted here as school districts, but only one district in a treated region receives a funding intervention. At the regional level, we only observe an aggregated treatment indicator and the aggregated outcome, not the specific district that received the intervention. The goal is to infer the high-resolution intervention locations from aggregated data.

The subregional context consists of high-resolution demographic and economic data, represented here by a normally distributed *wealth score*. We hypothesize that the effect of school funding depends on this wealth score through a global shift and scale.

Setup. We assume $N = 30 \times 30 = 900$ regions, each with $M = 2 \times 2 = 4$ subregions.

We simulate:

- A high-resolution intervention matrix $T \in \mathbb{R}^{N \times M}$, where $t_{i,j}$ indicates whether subregion j of region i receives the funding intervention.
- A regional intervention vector $\hat{T} \in \mathbb{R}^N$, where $\hat{T}_i = \sum_{j=1}^M t_{i,j}$ is the absolute intervention intensity in region i .
- A high-resolution context matrix $C \in \mathbb{R}^{N \times M}$ encoding the socioeconomic status of each subregion, where smaller $c_{i,j}$ values represent lower socioeconomic status and larger $c_{i,j}$ values represent higher socioeconomic status.
- A high-resolution noise matrix $\epsilon \in \mathbb{R}^{N \times M}$ with i.i.d. entries $\epsilon_{i,j} \sim \mathcal{N}(0, \sigma^2)$, where $\sigma^2 = 0.005$.

The intervention matrix T is generated by:

1. Initializing all entries to zero.
2. Randomly selecting 50% of the regions for treatment.
3. For each treated region i , choosing exactly one subregion $j \in \{1, \dots, M\}$ uniformly at random and assigning $t_{i,j} = 1$.

The context matrix C is sampled i.i.d. from a standard normal distribution:

$$c_{i,j} \sim \mathcal{N}(0, 1).$$

The outcome variable is given by the ground-truth functional relationship

$$y_{i,j} = f_{\theta}(t_{i,j}, c_{i,j}) + \epsilon_{i,j} = (\theta_{\text{shift}} - c_{i,j}) \cdot \theta_{\text{scale}} \cdot t_{i,j} + \epsilon_{i,j},$$

with ground-truth parameters

$$\theta_{\text{shift}} = 4.2, \quad \theta_{\text{scale}} = 1.23.$$

The observed regional outcome is aggregated by summation:

$$\hat{Y}_i = \sum_{j=1}^M y_{i,j}.$$

As in Exp. 1, we assume no hidden confounding, no time-series effects, treatment assignment independent of context, and aggregation-based regional observations.

Training. We train a model to estimate both the latent subregional intervention assignments and the parameters of the causal effect function.

The model has:

- $N \times M = 900 \times 4$ latent treatment logits, one vector of length M for each region.
- Two scalar parameters $\{\theta_{\text{shift}}, \theta_{\text{scale}}\}$ specifying the parametric form of $f_{\theta}(\cdot)$.

The learned causal effect function is parametrized as

$$f_{\theta}(t_{i,j}, c_{i,j}) = (\theta_{\text{shift}} - c_{i,j}) \cdot \theta_{\text{scale}} \cdot t_{i,j}.$$

To ensure a positive scale parameter, θ_{scale} is represented through a softplus transformation.

An auxiliary preprocessing function is applied to the latent treatment logits using the Gumbel-Softmax relaxation. This produces a soft row-wise assignment over the M subregions. The resulting assignment is processed so that untreated rows remain all-zero and treated rows have approximately one active subregion. During training, the Gumbel-Softmax temperature is exponentially annealed from 2.0 to 0.1.

In the forward pass:

1. Sample a mini-batch of region indices.
2. Process the latent treatment logits using the Gumbel-Softmax relaxation to obtain an estimate of the high-resolution treatment matrix T for the mini-batch.
3. Compute predicted subregional outcomes:

$$\mu_{i,j} = f_{\theta}(\hat{t}_{i,j}, c_{i,j}).$$

4. Aggregate to regional predictions by summation:

$$\mu_i = \sum_{j=1}^M \mu_{i,j}.$$

5. Compare the regional prediction with the observed regional outcome $\hat{Y}_i$ using the MSE loss.

We optimize all parameters jointly using AdamW with learning rate 10^{-4} . Training is performed for 200,000 epochs with a fixed random seed.

Details of Exp. 3: Spatiotemporal Effects of Heat Waves on School Performance

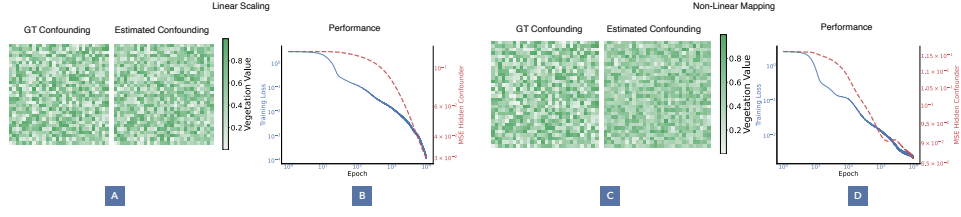


Figure 8: Ground-truth vegetation index, the corresponding estimated confounder fields, and joint plots of training loss and confounder MSE. Results are shown for two settings: linear scaling (left) and MLP learning (right).

Neural Network. Additionally, we train a neural-network variant following the same treatment-assignment procedure. In this version, the causal effect function $f_{\theta}(\cdot)$ is implemented as a feed-forward neural network taking $(t_{i,j}, c_{i,j})$ as input.

The network has three hidden layers with hidden dimension 16 and tanh activations:

$$(t_{i,j}, c_{i,j}) \mapsto \text{MLP}_{\theta}(t_{i,j}, c_{i,j}).$$

To stabilize the local-effect interpretation, the model uses the centered form

$$f_{\theta}(t_{i,j}, c_{i,j}) = \text{MLP}_{\theta}(t_{i,j}, c_{i,j}) - \text{MLP}_{\theta}(0, c_{i,j}),$$

which enforces $f_{\theta}(0, c_{i,j}) \approx 0$. The latent treatment indicators are processed in the same way as in the parametric model, using Gumbel-Softmax with temperature annealing from 2.0 to 0.1.

The neural-network model is also trained for 200,000 epochs using AdamW with learning rate 10^{-4} and mini-batches containing 20% of the regions.

Results. Figure 7a shows the evolution of the mini-batch training loss and the quality of the treatment-location reconstruction (measured as MAE). Both the neural network and the parametric function recover the locations reasonably well. We however find that this depends on the noise level and on proper temperature annealing. The final reconstructed locations are shown in Figure 7b, where incorrect predictions are shown in purple.

D.3 Exp. 3: Spatiotemporal Effects of Heat Waves on School Performance

Context. In this experiment, we examine the impact of extreme heat events (“heat waves”) on educational outcomes across both space and time. We model a spatial grid of $N = 100$ regions, each containing $M = 9$ subregions, observed over $W = 48$ consecutive months. The intervention represents the occurrence of a heat wave in a given subregion and month. Heat waves have a detrimental effect on school performance, but the strength of this effect depends on the educational background of parents in the subregion and on an unobserved environmental factor, vegetation coverage, which mitigates the impact of heat.

The observed context for each subregion is a categorical indicator of parents’ education level, taking one of three values (*low* = 1, *medium* = 2, *high* = 3), which evolves slowly over time to reflect gradual demographic changes. The unobserved confounding context is a fixed vegetation index between 0 and 1 for each subregion, representing the proportion of green cover and its protective effect against heat. The intervention process is binary at the subregional level, with most regions remaining unaffected in a given month, but some experiencing heat waves simultaneously in all their subregions, corresponding to large-scale weather patterns.

The outcome, measured as a monthly change in school performance for each subregion, is driven by the interaction between the intervention, parents’ education level, and the vegetation coverage. Subregions with lower parental education are more strongly affected, while vegetation dampens the negative effect. The challenge in this experiment is to recover both the mapping from context, intervention, and unobserved confounder to outcomes, and the vegetation values themselves, given only aggregated regional-level outcomes over time.

Setup. We model $N = 100$ regions, each with $M = 9$ subregions, observed over $W = 48$ discrete time points (months).

For each month $w \in \{1, \dots, W\}$, we define:

- A high-resolution binary treatment matrix $T^{(w)} \in \{0, 1\}^{N \times M}$, where $t_{i,j}^{(w)} = 1$ indicates that subregion j of region i is experiencing a heat wave in month w . In our data generation, each row of $T^{(w)}$ is either all zeros (probability 0.7) or all ones (probability 0.3), corresponding to large-scale heat events affecting entire regions.
- An observed context matrix $C^{(w)} \in \{1, 2, 3\}^{N \times M}$ encoding the categorical education level of parents in each subregion. At $w = 1$, these values are sampled uniformly. For $w > 1$, $C^{(w)}$ is obtained from $C^{(w-1)}$ by flipping each entry to a random category with probability 0.05, capturing slow demographic change.
- An unobserved covariate context matrix $U \in [0, 1]^{N \times M}$ containing fixed vegetation index values for each subregion, drawn uniformly from $[0, 1]$ and constant across all months.
- A noise matrix $\epsilon^{(w)} \in \mathbb{R}^{N \times M}$ with i.i.d. entries $\epsilon_{i,j}^{(w)} \sim \mathcal{N}(0, \sigma^2)$, with $\sigma^2 = 0.02$.

The subregional outcome is generated as:

$$y_{i,j}^{(w)} = f_{\theta}(t_{i,j}^{(w)}, c_{i,j}^{(w)}, u_{i,j}) + \epsilon_{i,j}^{(w)},$$

where $c_{i,j}^{(w)}$ is the parents' education category, $u_{i,j}$ is the vegetation index, and:

$$f_{\theta}(t_{i,j}^{(w)}, c_{i,j}^{(w)}, u_{i,j}) = \begin{cases} 0 & \text{if } t_{i,j}^{(w)} = 0, \\ (10 \cdot \mathbb{1}[c_{i,j}^{(w)} = 1] + 5 \cdot \mathbb{1}[c_{i,j}^{(w)} = 2] + \mathbb{1}[c_{i,j}^{(w)} = 3]) \cdot (1 - u_{i,j}) & \text{if } t_{i,j}^{(w)} = 1. \end{cases}$$

This function specifies that the subregional outcome depends on whether a heat wave occurs and, if so, how vulnerable the subregion is based on the education level of parents and the vegetation coverage. If $t_{i,j}^{(w)} = 0$, the effect is zero. If $t_{i,j}^{(w)} = 1$, the impact is determined by the categorical education level with coefficients (10, 5, 1) corresponding to low, medium, and high parental education, respectively, and is scaled by $(1 - u_{i,j})$, which reduces the effect in proportion to the amount of vegetation present. The indicator $\mathbb{1}[\cdot]$ selects the appropriate coefficient for each subregion.

The regional outcome at month w is obtained via mean aggregation. This setup induces a spatiotemporal causal problem with heterogeneous treatment effects driven by both observed and unobserved context, where the latter must be recovered from aggregated outcomes across multiple time points.

Training. We train a model to jointly estimate:

1. The mapping $f_{\theta}(\cdot)$ from observed context, intervention, and vegetation index to subregional outcomes,
2. The static vegetation index values $U \in [0, 1]^{N \times M}$ for all subregions.

The function $f_{\theta}(\cdot)$ is implemented as a 4 layer multilayer perceptron (MLP) with hidden dimension 16 and a dropout rate of 0.1 after the second hidden layer. Its inputs for each subregion are:

$$z_{i,j}^{(w)} = (\mathbb{1}[c_{i,j}^{(w)} = 1], \mathbb{1}[c_{i,j}^{(w)} = 2], \mathbb{1}[c_{i,j}^{(w)} = 3], t_{i,j}^{(w)}, u_{i,j}),$$

where $u_{i,j}$ is the vegetation index. The output is the predicted outcome $y_{i,j}^{(w)}$ for that subregion and month.

In the forward pass for a given month w :

1. The model takes as input $T^{(w)}$, $C^{(w)}$, and U ,
2. The MLP evaluates $f_{\theta}(\cdot)$ at each (i, j) to produce predicted subregional outcomes $y_{i,j}^{(w)}$,

3. These are aggregated to the regional level via mean aggregation.
4. The primary loss is the MSE between predicted and observed regional outcomes.

Training proceeds over all months in random order at each epoch. The total parameter set consists of the MLP weights θ and the vegetation matrix U . We optimize using Adam with a learning rate of 0.001 for 10,000 epochs, with fixed random seeds for reproducibility. Loss curves for $\mathcal{L}_{\text{region}}$ and the MSE of $\mathcal{L}_{\text{veg}}$ are recorded throughout training.

Results.

1. Figure 8a shows the loss curves and vegetation MSE under the linear scaling parameterization. Both training loss and MSE decrease rapidly, and the final error is very small, indicating near-perfect recovery of the unobserved covariate.
2. Figure 8b compares the ground-truth vegetation field to the estimated vegetation under linear scaling. The two maps are almost indistinguishable, confirming that the restricted functional form matches the data-generating process and allows highly accurate recovery.
3. Figures 8c and d present the same results for the non-linear MLP parameterization. While the model still recovers meaningful structure in the vegetation, the MSE remains orders of magnitude higher than in the linear case, and deviations from the ground truth are clearly visible in the estimated field. This is notable since the true vegetation values are not necessarily identifiable; the MLP could instead store a transformed version and map them back during the forward process.
4. Overall, these results demonstrate that the model can reconstruct regional outcomes while also uncovering hidden drivers of heterogeneity. The comparison between linear and non-linear parameterizations highlights the trade-off: structural restrictions yield more accurate recovery when they match the true mechanism, while flexible approximators such as MLPs still learn the vegetation but with reduced precision.

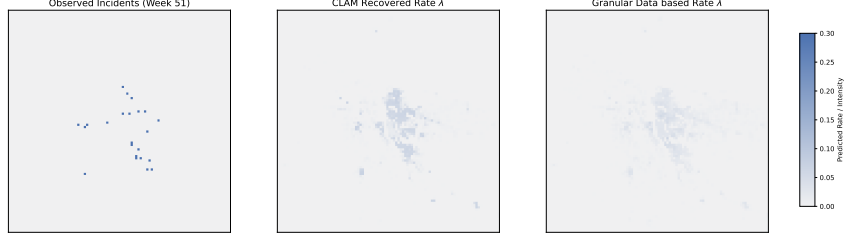


Figure 9: Comparison of learned spatial intensities for week 51. Both models are rendered on a shared color scale across the $\sqrt{N} \cdot M \times \sqrt{N} \cdot M$ grid, alongside the observed events for that week.

E Real-World Case Study Details: Effect of Heat on Gun Violence

This section provides the full details of the real-world study summarized in Section 6.2. The notation follows Section 2: regions are indexed by $i \in \{1, \dots, N\}$, subregions or cells by $j \in \{1, \dots, M\}$, and weeks by $w \in \{1, \dots, W\}$. Temperature is used as a continuous, time-varying treatment-like exposure; urbanization is the HR static contextual covariate; and gun-incident counts are the observed LR outcomes.

Problem Setting. The target variable is observed at a coarse spatial resolution, while covariates are available at a finer resolution. The study area is discretized into a fine grid of $\sqrt{N} \cdot M \times \sqrt{N} \cdot M$ cells, grouped into $\sqrt{N} \times \sqrt{N}$ coarse regions. With $N = 100$ and $M = 100$, this gives a 100×100 cell grid grouped into a 10×10 region grid. For each week w and region i , we observe a nonnegative count outcome

$$Y_{i,w}^{\text{LR}} \in \mathbb{Z}_{\geq 0}, \quad (5)$$

representing the number of gun-incident events occurring in region i during week w .

Data Representation and Covariates. Let $c_{i,j,w}$ denote the observed urbanization measurement for cell (i, j) and week w . Since urbanization is treated as time-invariant in this study, we use its temporal mean,

$$c_{i,j} = \frac{1}{W} \sum_{w=1}^W c_{i,j,w}. \quad (6)$$

Temperature is modeled at the region level. Let $t_{i,j,w}^{\text{HR}}$ denote the cell-level temperature for cell (i, j) and week w . We compute the region-average temperature exposure as

$$t_{i,w}^{\text{LR}} = \frac{1}{M} \sum_{j=1}^M t_{i,j,w}^{\text{HR}}, \quad (7)$$

and standardize it using the global mean and standard deviation across all region-week pairs:

$$\tilde{t}_{i,w} = \frac{t_{i,w}^{\text{LR}} - \mu_t}{\sigma_t + \varepsilon}, \quad (8)$$

where $\varepsilon > 0$ is a small constant for numerical stability. Missing temperature values are imputed using the global median temperature, and missing outcome counts are set to zero.

Disaggregation Model. We model the latent HR intensity, or rate, $\lambda_{i,j,w} > 0$ as a function of the standardized regional temperature exposure and the HR urbanization covariate:

$$\lambda_{i,j,w} = f_{\theta}(\tilde{t}_{i,w}, c_{i,j}), \quad (9)$$

where $f_{\theta}(\cdot)$ is a feed-forward neural network with parameters θ . To enforce nonnegativity, the network output is passed through a softplus transformation.

The model-implied LR expected count is obtained by summing the HR rates over all cells in region i :

$$\mu_{i,w} = \sum_{j=1}^M \lambda_{i,j,w}. \quad (10)$$

Thus, $\mu_{i,w}$ denotes the predicted regional count intensity, while $Y_{i,w}^{\text{LR}}$ denotes the observed regional count. This additive aggregation is consistent with the SCM in Section 3.1.

Training Objective. We assume a Poisson observation model at the region–week level,

$$Y_{i,w}^{\text{LR}} \sim \text{Poisson}(\mu_{i,w}), \quad (11)$$

and train f_θ by minimizing the negative Poisson log-likelihood, up to additive constants:

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{W}_{\text{train}}| \cdot N} \sum_{w \in \mathcal{W}_{\text{train}}} \sum_{i=1}^N (\mu_{i,w} - Y_{i,w}^{\text{LR}} \log(\mu_{i,w} + \varepsilon)), \quad (12)$$

where $\mathcal{W}_{\text{train}}$ is the set of training weeks and $\varepsilon > 0$ is a small constant for numerical stability.

Spatial Intensity Comparison. Figure 9 visualizes observed incidents for a representative week, week 51, alongside spatial intensity estimates learned under different supervision regimes. When trained only on region-level counts, CLAM produces spatially structured intensity maps that concentrate mass around incident clusters but exhibit noticeably higher peak intensities and broader spatial support compared to the cell-supervised estimate. This overestimation reflects a fundamental ambiguity induced by aggregation: without access to fine-grained outcomes, the model must allocate regional mass across plausible subregional locations, leading to inflated local rates in areas consistent with the observed context and covariates. The response-surface comparison in the main paper, Figure 4, shows the same pattern in covariate space rather than physical space.

Learning. The training procedure optimizes the function parameters θ (or any other missing components of the SCM). The structural function $f_\theta(\cdot)$ can be represented by a neural network or any other parameterized function. Since the fine-grained outcomes $y_{i,j}$ are unobserved, training relies solely on their aggregated counterparts.

We optimize with the goal that the inferred aggregated predictions (μ_i) match the observed aggregate: $\mathcal{L}_{\theta,\phi} = \sum_{i=1}^N (\hat{Y}_i - \mu_i)^2$, where $\hat{Y}_i$ denotes the observed aggregate of region i . If the noise terms are not i.i.d. and normally distributed, they can also be explicitly learned within the optimization loop.

Details of Exp. 4: Aggregation Function Estimation

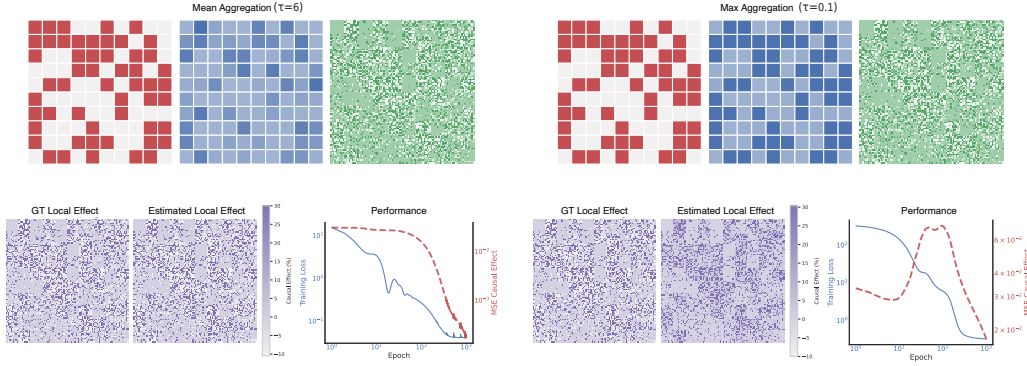


Figure 10: **Results Overview.** **Top:** Input of LR intervention locations (red), LR outcomes (blue), and HR context (green) for mean aggregation and max aggregation cases. **Bottom:** We observe the estimated local effects and the performance of our model over multiple epochs for both cases.

F Additional Experiments

Exp. 4 considers driving bans with an unknown aggregation function between subregional and regional outcomes. Exp. 5 introduces treatment–context confounding, where the probability of subregional treatment depends on contextual variables, affecting both allocation and outcomes

F.1 Exp. 4: Unknown Aggregation Functions

Context. We study the causal effect of driving bans on air quality under the assumption that the reported region-level air quality values are an unknown aggregation of finer-scale (subregional) measurements. Specifically, we model the aggregation as a learned combination of the subregional mean and maximum values, with the combination weights parameterized via a logit transform.

A realistic example is a city-wide driving ban where air quality is officially reported as a single regional index (e.g., $\text{PM}_{2.5}$ concentration), while the underlying measurement network collects data from multiple monitoring stations. Depending on reporting policies, the published index might resemble an average across stations, a worst-case station reading, or something in between.

The high-resolution context for each subregion is a single continuous variable: a vegetation index, capturing the amount of green coverage in the area. Vegetation can mitigate pollution and thus may modulate the effect of a driving ban. We hypothesize that the causal effect of driving bans varies with vegetation and that the unknown aggregation mechanism must be learned alongside the effect parameters to estimate subregional impacts correctly.

Setup. We assume $N = 100$ regions, each with $M = 100$ subregions.

We are given:

- A binary treatment vector $T \in \{0, 1\}^N$, where $T_i = 1$ indicates that a driving ban was implemented in region i (and thus all its subregions).
- A high-resolution context matrix $C \in [0, 1]^{N \times M}$ encoding the vegetation index of each subregion, where $c_{i,j} = 0$ represents minimal vegetation cover and $c_{i,j} = 1$ represents dense vegetation.
- A high-resolution noise matrix $\epsilon \in \mathbb{R}^{N \times M}$ with i.i.d. entries $\epsilon_{i,j} \sim \mathcal{N}(0, \sigma^2)$, where $\sigma^2 = 0.02$.

The treatment vector T is generated using a Bernoulli distribution with probability 0.5, assigning each region to either the intervention or control group at random. The vegetation index values in C are drawn i.i.d. from $\text{Uniform}(0, 1)$.

The subregional outcome variable is given by the ground-truth functional relationship:

$$y_{i,j} = f_{\theta}(t_{i,j}, c_{i,j}) + \epsilon_{i,j},$$

where:

- $t_{i,j} = T_i$ is the treatment status of subregion j in region i (inherited from the region),
- $c_{i,j}$ is the vegetation index,
- $\epsilon_{i,j}$ is Gaussian noise.

The causal effect function is parameterized as:

$$f_{\theta}(t_{i,j}, c_{i,j}) = \begin{cases} 0.0 & \text{if } t_{i,j} = 0, \\ \theta_{\text{base}} + \theta_{\text{veg}} \cdot c_{i,j} & \text{if } t_{i,j} = 1, \end{cases}$$

where θ_{base} captures the baseline effect of a driving ban and θ_{veg} captures how this effect changes with vegetation cover.

Unlike Exp. 1, the regional outcome x_i is not a simple mean over subregions. Instead, we model the aggregation function as:

$$y_i = \sum_{j=1}^M p_{i,j}(\tau) \cdot y_{i,j},$$

where:

$$p_{i,j}(\tau) = \frac{\exp(y_{i,j}/\tau)}{\sum_{k=1}^M \exp(y_{i,k}/\tau)}$$

is the softmax weight assigned to subregion j with temperature parameter $\tau > 0$. When $\tau \rightarrow \infty$, $p_{i,j}$ approaches a uniform distribution (mean aggregation); when $\tau \rightarrow 0$, $p_{i,j}$ concentrates on the subregion with the highest outcome (max aggregation).

Training. We train a model to estimate:

1. The parameters $\theta = \{\theta_{\text{base}}, \theta_{\text{veg}}\}$ of the causal effect function $f_{\theta}(\cdot)$,
2. The aggregation temperature parameter $\tau > 0$ governing the softmax-based aggregation from subregional to regional outcomes.

The learned function takes the form:

$$f_{\theta}(t_{i,j}, c_{i,j}) = \begin{cases} 0.0 & \text{if } t_{i,j} = 0, \\ \theta_{\text{base}} + \theta_{\text{veg}} \cdot c_{i,j} & \text{if } t_{i,j} = 1. \end{cases}$$

In the forward pass:

1. Compute predicted subregional outcomes: $y_{i,j} = f_{\theta}(t_{i,j}, c_{i,j})$,
2. Compute subregion-level aggregation weights via the softmax: $p_{i,j}(\tau) = \frac{\exp(y_{i,j}/\tau)}{\sum_{k=1}^M \exp(y_{i,k}/\tau)}$,
3. Compute predicted regional outcomes as the expectation under $p_{i,j}(\tau)$: $y_i = \sum_{j=1}^M p_{i,j}(\tau) \cdot y_{i,j}$,
4. Compute the loss as the mean squared error with the observed Y_i .

All parameters $\{\theta_{\text{base}}, \theta_{\text{veg}}, \tau\}$ are optimized jointly using Adam with a learning rate of 0.001 for 1000 epochs. We initialize τ to a moderate value (e.g., $\tau = 1.0$) and enforce $\tau > 0$ during training by optimizing its logarithm.

Results.

1. Visualizations of the aggregated causal effects show that mean aggregation preserves more variation across regions, whereas max aggregation suppresses smaller effects and produces sharper contrasts.
2. Although the subregional causal effect loss is never used directly for training, it decreases steadily over epochs, indicating that the model recovers fine scale effects from only regional supervision. Training loss is substantially higher under max aggregation, as weaker subregional signals are obscured.

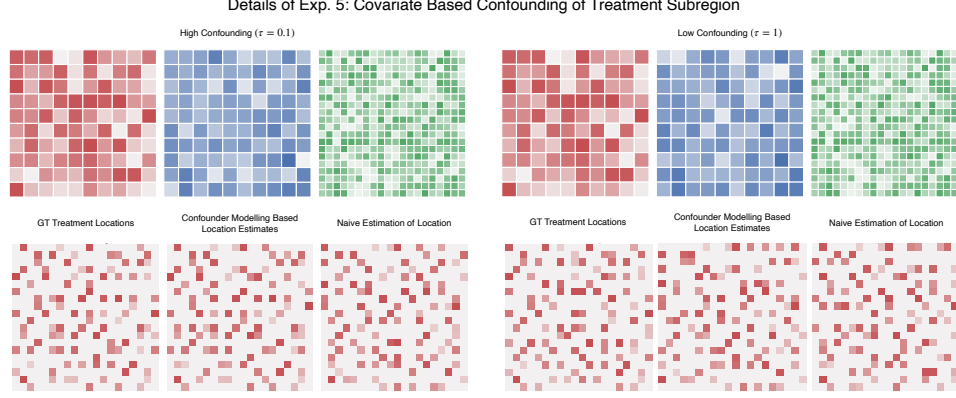


Figure 11: **Results Overview.** **Top:** Input of LR intervention locations (red), LR outcomes (blue), and HR context (green). **Bottom:** Ground truth treatment locations, estimation of treatment locations based on modeling of location covariate-based confounding, naïve estimation of treatment locations with all free parameters.

3. Comparing estimated and ground truth local effects confirms this bias: areas with small true effects tend to be overestimated when max aggregation dominates.
4. The aggregation function, parameterized by the temperature τ , is recovered with high accuracy. In the mean aggregation case, τ is estimated within 0.01 of the ground truth, while in the max aggregation case, the estimate is slightly underestimated (5.6), consistent with the plateauing behavior of the softmax function at low temperatures.
5. Overall, the experiment demonstrates that our approach can successfully recover both the aggregation mechanism (within its parametric form) and the underlying causal effects, despite only observing region-level outcomes.

F.2 Exp. 5: Covariate-Based Confounding of Treatment Subregion

Context. In this experiment, we study how confounding between treatment allocation and contextual variables affects causal effect estimation. We take the example of school districts receiving additional public funding and the corresponding changes in educational outcomes.

In reality, funding allocation is rarely random: wealthier or poorer districts may have systematically different chances of receiving funding due to political priorities, lobbying power, or policy constraints. This creates a dependency between the probability of treatment and district level context variables, violating the standard assumption of independent treatment assignment.

We model this confounding explicitly by making the probability of treatment allocation a deterministic function of the contextual variable, passed through a logistic transform to obtain valid probabilities for each region. The treatment effect itself is also a function of the same contextual variable, representing heterogeneous impacts across different socioeconomic conditions.

Setup. We assume $N = 100$ regions, each with $M = 100$ subregions (school districts).

We are given:

- A high-resolution treatment matrix $T \in \{0, 1\}^{N \times M}$ with *exactly one* treated subregion per region, i.e., $\sum_{j=1}^M t_{i,j} = 1$ for all i .
- A context matrix $C \in [0, 1]^{N \times M}$ encoding a socioeconomic/need index for each subregion, where a higher $c_{i,j}$ indicates greater need.
- A noise matrix $\epsilon \in \mathbb{R}^{N \times M}$ with i.i.d. entries $\epsilon_{i,j} \sim \mathcal{N}(0, \sigma^2)$ (default $\sigma^2 = 0.02$).

Confounded Treatment Allocation. For each region i , we form logits from the subregional context,

$$\ell_{i,j} = \alpha_0 + \alpha_1 c_{i,j},$$

and convert them to a categorical distribution over subregions via a row-wise softmax,

$$p_{i,j} = \frac{\exp(\ell_{i,j})}{\sum_{k=1}^M \exp(\ell_{i,k})}.$$

We then draw exactly one treated subregion $j^* \sim \text{Categorical}(p_{i,1:M})$ and set

$$t_{i,j} = \mathbb{1}\{j = j^*\}, \quad \sum_{j=1}^M t_{i,j} = 1.$$

The parameters (α_0, α_1) control the strength and direction of confounding between context and treatment assignment (default $\alpha_0 = 0$, $\alpha_1 > 0$ so higher-need subregions are more likely to be treated).

Outcome Model With Context-Dependent Effects. Subregional outcomes follow the ground-truth functional relationship

$$y_{i,j} = f_{\theta}(t_{i,j}, c_{i,j}) + \epsilon_{i,j},$$

with the causal effect function

$$f_{\theta}(t_{i,j}, c_{i,j}) = \begin{cases} 0 & \text{if } t_{i,j} = 0, \\ \theta_{\text{base}} + \theta_{\text{soc}} c_{i,j} & \text{if } t_{i,j} = 1, \end{cases}$$

where $\theta = \{\theta_{\text{base}}, \theta_{\text{soc}}\}$ captures the baseline funding effect and its modulation by socioeconomic need.

Regional Aggregation. Region-level outcomes are the sum over subregions (as in Exp. 1).

This construction induces confounding because (i) treatment allocation depends on $c_{i,j}$ through the softmax logits, and (ii) treatment effects also vary with $c_{i,j}$ via $f_{\theta}(\cdot)$.

Training. We compare two training regimes for estimating the causal effect parameters $\theta = \{\theta_{\text{base}}, \theta_{\text{soc}}\}$ and the high resolution treatment assignments $T \in \mathbb{R}^{N \times M}$ in the presence of treatment context confounding.

Regime 1: Naïve Training Under Independence Assumption. This setup directly applies the training procedure of Exp. 2 (Section D) to the confounded data, it:

1. Treats the treatment allocation as if independent of the context,
2. Learns T as free parameters, constrained to be one hot per row via a differentiable argmax with temperature,
3. Learns θ by minimizing the MSE between observed and predicted regional outcomes.

This ignores the fact that the treatment assignment mechanism is context dependent.

Regime 2: Confounding Aware Training. Here, we explicitly model the context-dependent treatment allocation mechanism. For each region i , we predict logits

$$\ell_{i,j} = \beta_0 + \beta_1 c_{i,j},$$

where (β_0, β_1) are learned parameters, and convert them to a categorical distribution via a row-wise softmax with a learnable temperature $\tau_{\text{conf}} > 0$:

$$p_{i,j}(\tau_{\text{conf}}) = \frac{\exp(\ell_{i,j}/\tau_{\text{conf}})}{\sum_{k=1}^M \exp(\ell_{i,k}/\tau_{\text{conf}})}.$$

We then use $p_{i,j}(\tau_{\text{conf}})$ as the (differentiable) treatment assignment in the forward pass.

In both regimes, predicted subregional outcomes are given by:

$$y_{i,j} = f_{\theta}(t_{i,j}, c_{i,j}) = \begin{cases} 0 & \text{if } t_{i,j} = 0, \\ \theta_{\text{base}} + \theta_{\text{soc}} \cdot c_{i,j} & \text{if } t_{i,j} = 1, \end{cases}$$

where in Regime 1, $t_{i,j}$ comes from the naïvely learned $\hat{T}$, and in Regime 2, $t_{i,j} = p_{i,j}(\tau_{\text{conf}})$ from the confounding model.

Regional outcomes are summed and aggregated. and the training loss is the mean squared error, All parameters are optimized jointly using Adam with a learning rate of 0.001 for 1000 epochs. Regime 1 learns $\{\hat{T}, \theta_{\text{base}}, \theta_{\text{soc}}\}$, while Regime 2 learns $\{\beta_0, \beta_1, \tau_{\text{conf}}, \theta_{\text{base}}, \theta_{\text{soc}}\}$.

Results.

1. We compare two cases of confounding in subregional treatment allocation, holding region-level treatments and contextual factors fixed. The aggregated causal effects differ markedly depending on the level of confounding, showing that the treatment context dependency directly shapes the observed regional outcomes.
2. We visualize the ground truth intervention locations alongside the estimates obtained with and without explicitly modeling confounding. Under strong confounding, the confounding-aware approach recovers the true intervention locations with high fidelity, while the naïve method from Exp. 2 retrieves only a small fraction correctly.
3. In the low confounding setting, both methods perform similarly, and location probabilities are estimated with comparable accuracy. This confirms that adjusting for confounding does not reduce performance when confounding is weak.
4. For causal effect estimation, the final MSE of subregional effects under low confounding was 2.04 (naïve) versus 1.60 (confounding aware). Under high confounding, the respective errors were 0.74 versus 0.32. These results show that accounting for confounding improves both intervention location recovery and the accuracy of estimated causal effects by conditioning on the correct contextual information.

Overall, this experiment demonstrates that explicitly modeling treatment context confounding is essential as it enables accurate recovery of intervention locations and yields more reliable causal effect estimates, particularly when treatment assignment is strongly biased by contextual factors.